

U20ECCM04

INTERNET OF THINGS
(Common to ECE, CCE and ICE)

L	T	P	C	Hrs
3	0	0	3	45

Course Objectives

- To impart necessary and practical knowledge of components of Internet of Things.
- To attain the knowledge about different types of Reference modules and architecture of IoT.
- To understand the concepts of Hardware and Software Elements.
- To acquire the knowledge about various Functions with IoT elements.
- To develop skills required to build real-time IoT based Applications.

Course Outcomes

After completion of the course, students will be able to

CO1- Infer internet of Things and its components.(K2)

CO2- Describe about Reference modules and Architecture.(K2)

CO3- Explain the concepts of Hardware and Software Elements .(K2)

CO4- Build and deploy various Functions with IoT elements.(K3)

CO5- Develop real-time IoT based Applications.(K3)

UNIT – I INTRODUCTION TO INTERNET OF THINGS (9 Hrs)

The Internet of Things Today, Time for Convergence, Towards the IoT Universe, Internet of Things Vision, IoT Strategic Research and Innovation Directions, IoT Applications, Future Internet Technologies, Infrastructure, Networks and Communication, Processes, Data Management, Security, Privacy & Trust.

UNIT - II ARCHITECTURE OF IoT (9 Hrs)

State of the Art – Introduction, Architecture Reference Model- IoT reference Model, IoT Reference Architecture, Functional View, Information View, Deployment and Operational View, Other Relevant architectural views.

UNIT - III ELEMENTS OF IoT (9 Hrs)

Hardware Components- Computing (Arduino, Raspberry Pi), Communication, Sensing, Actuation, I/O interfaces.

Software Components- Programming APIs (using Python/Node.js/Arduino) for Communication Protocols-MQTT, ZigBee, Bluetooth, CoAP, UDP, TCP.

UNIT - IV IoT DEVELOPMENT (9 Hrs)

Solution framework for IoT applications- Implementation of Device Integration, Data acquisition, and integration, Device data storage- Unstructured data storage on cloud/local server, Authentication, authorization of devices

UNIT -V IoT APPLICATIONS (9 Hrs)

IoT applications for industry: Future Factory Concepts, Brownfield IoT, Smart Objects, Smart Applications, Four Aspects in Business to Master IoT, IoT for Retailing Industry, IoT for Oil and Gas Industry.

Text Books

1. Vijay Madiseti, Arshdeep Bahga, Internet of Things, "A Hands-on Approach", University Press,3rd/e, Aug 2018.
2. Raj Kamal, "Internet of Things: Architecture and Design", McGraw Hill ISBN: 9789352605224, 9789352605224,2nd edition, May 2017
3. Dr. SRN Reddy, Rachit Thukral and Manasi Mishra, "Introduction to Internet of Things: A Practical Approach", ETI Labs 2014

Reference Books

1. Jeeva Jose, "Internet of Things", Khanna Publishing House, Delhi, 2012
2. Adrian McEwen, "Designing the Internet of Things", Wiley, 2007

B.Tech-Computer and Communication Engineering

V. Bhanu

3. Raj Kamal, "Internet of Things: Architecture and Design", McGraw Hill, 2002
4. Cuno Pfister, "Getting Started with the Internet of Things", O Reilly Media, 2015
5. Pethuru Raj and Anupama C. Raman, "The Internet of Things: Enabling Technologies, Platforms, and Use Cases", CRC Press

Web References

1. <https://www.i-scoop.eu/internet-of-things-guide/>
2. <https://www.theinternetofthings.eu/>
3. <https://www.udemy.com/course/complete-guide-to-build-iot-things-from-scratch-to-market/>
4. <https://www.coursera.org/learn/iot>
5. https://onlinecourses.nptel.ac.in/noc21_ee85/preview

COs/POs/PSOs Mapping

COs	Program Outcomes (POs)												Program Specific Outcomes (PSOs)		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	2	2	3	2	-	-	-	-	-	-	-	-	3	3	
2	3	-	3	2	-	-	-	-	-	-	-	-	3	3	
3	2	3	2	-	-	-	-	-	-	-	-	-	3	3	
4	2	2	2	-	-	-	-	-	-	-	3	-	3	3	1
5	2	3	2	-	3	-	-	-	-	-	3	-	3	3	2

Correlation Level: 1-Low, 2-Medium, 3-High

U20CCT614	INTERNET PROGRAMMING	L	T	P	C	Hrs
		3	0	0	3	45

Course Objectives

- To understand different Internet Technologies.
- To learn java-specific client side programming.
- To learn java servlet programming and database connectivity.
- To learn PHP and XML document.
- Understand AJAX and services provided by web.

Course Outcomes

- CO1-** Construct a basic website using HTML and Cascading Style Sheets. **(K2)**
- CO2-** Build dynamic web page with validation using Java Script objects and by applying different event handling mechanisms. **(K3)**
- CO3-** Develop server side programs using Servlets and JSP. **(K3)**
- CO4-** Construct simple web pages in PHP and to represent data in XML format. **(K3)**
- CO5-** Use AJAX and web services to develop interactive web applications. **(K3)**

UNIT I INTRODUCTION**(9 Hrs)**

Web Essentials: Clients, Servers and Communication – The Internet – Basic Internet protocols – World wide web – HTTP Request and Response – Web Clients – Web Servers – HTML5 – Tables – Lists – Image – HTML5 control elements – Semantic elements – Drag and Drop – Audio – Video controls – CSS3 – Inline, embedded and external style sheets – Rule cascading – Inheritance – Backgrounds – Border Images – Colors – Shadows – Text – Transformations – Transitions – Animations.

UNIT II CLIENT SIDE PROGRAMMING**(9 Hrs)**

Java Script: An introduction to JavaScript–JavaScript DOM Model–Date and Objects, Regular Expressions– Exception Handling–Validation–Built-in objects–Event Handling DHTML with JavaScript–JSON introduction – Syntax – Function Files – Http Request – SQL.

UNIT III SERVER SIDE PROGRAMMING**(9 Hrs)**

Servlets: Java Servlet Architecture- Servlet Life Cycle- Form GET and POST actions, Session Handling- Understanding Cookies- Installing and Configuring Apache Tomcat Web Server

Database Connectivity: JDBC perspectives, JDBC program example - JSP: Understanding Java Server Pages-JSP Standard Tag Library (JSTL)-Creating HTML forms by embedding JSP code.

UNIT IV PHP and XML**(9 Hrs)**

An introduction to PHP: PHP- Using PHP- Variables- Program control- Built-in functions, Form Validation- Regular Expressions - File handling – Cookies - Connecting to Database.

XML: Basic XML- Document Type Definition- XML Schema DOM and Presenting XML, XML Parsers and Validation, XSL and XSLT Transformation, News Feed (RSS and ATOM).

UNIT V INTRODUCTION TO AJAX and WEB SERVICES**(9 Hrs)**

AJAX: Ajax Client Server Architecture-XML Http Request Object-Call Back Methods.

Web Services: Introduction- Java web services Basics – Creating, Publishing, Testing and Describing a Web services (WSDL)-Consuming a web service, Database Driven web service from an application –SOAP.

Text Books

1. Deitel and Deitel and Nieto, "Internet and World Wide Web How to Program", Pearson Education, 5th Edition, 2011.
2. Robert W. Sebesta, "Programming with World Wide Web", Pearson Education, Eighth Edition, 2015.
3. Jeffrey C and Jackson, "Web Technologies A Computer Science Perspective", Pearson Education, 2011.

Reference Books

1. Stephen Wynkoop and John Burke, "Running a Perfect Website", QUE, 2nd Edition, 1999.
2. Chris Bates, Web Programming, "Building Intranet Applications", Wiley Publications, 3rd Edition, 2009.
3. Gopalan N.P. and Akilandeswari J., "Web Technology", Prentice Hall of India, 2011.
4. Uttam K. Roy, "Web Technologies", Oxford University Press, 2011.
5. Kogent Learning Solutions Inc., Web Technologies: Black Book, Dreamtech publication, 2009.

Web References

1. [www. w3schools.com](http://www.w3schools.com)
2. <https://www.javatpoint.com>
3. <https://www.tutorialspoint.com>

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2	3	2	1	1	1	-	-	-	-	-	1	1	2	3	-
3	3	2	1	1	1	-	-	-	-	-	1	1	2	3	-
4	3	2	1	1	-	-	-	-	-	-	1	1	2	3	-
5	3	2	1	1	-	-	-	-	-	-	1	1	2	3	-

Correlation Level: 1-Low, 2-Medium, 3- High

U20CCT615	WIRELESS COMMUNICATION SYSTEMS	L	T	P	C	Hrs
		3	0	0	3	45

Course Objectives

- To learn about evolution of mobile communication technology.
- To understand the basic concepts and terms of cellular system.
- To describe the various medium access control used in wireless communication.
- To learn CDMA based communication systems
- To understand recent wireless technologies.

Course Outcomes

After completion of the course, the students are able to

CO1- Explain development in mobile communication. **(K2)**

CO2- Demonstrate the concepts of cellular system, resource availability and traffic demands. **(K2)**

CO3- Illustrate the various medium access alternatives for wireless communication. **(K2)**

CO4- Analyze performance of CDMA based system. **(K3)**

CO5- Explain the principles of recent wireless communication. **(K2)**

UNIT I INTRODUCTION**(9 Hrs)**

Evolution of mobile communications, Mobile Radio System around the world, Types of Wireless communication System, Comparison of Common wireless system, Trend in Cellular radio and personal communication. Second generation Cellular Networks, Third Generation (3G) Wireless Networks, Wireless Local Loop(WLL), Wireless Local Area network(WLAN), Bluetooth and Personal Area Networks

UNIT II CELLULAR CONCEPT AND SYSTEM DESIGN FUNDAMENTALS**(9 Hrs)**

Cellular system, Hexagonal geometry cell and concept of frequency reuse, Channel Assignment Strategies Distance to frequency reuse ratio, Channel & co-channel interference reduction factor, S/I ratio consideration and calculation for Minimum Co-channel and adjacent interference, Handoff Strategies, Umbrella Cell Concept, Trunking and Grade of Service, Improving Coverage & Capacity in Cellular System-cell splitting, Cell sectorization, Repeaters, Micro cell zone concept

UNIT –II MEDIUM ACCESS ALTERNATIVES FOR WIRELESS COMMUNICATION (9 Hrs)

Spread Spectrum Modulation - Pseudo-Noise Codes with Properties and Code Generation Mechanisms -DSSS and FHSS Systems - Time Hopping and Hybrid Spread Systems; Multicarrier Modulation Techniques -Zero Inter Symbol Interference Communication Techniques - Detection Strategies - Diversity Combining Techniques: Selection Combining - Threshold Combining - Equal Gain Combining - Maximum Ratio Combining.

UNIT IV WIRELESS SYSTEMS**(9 Hrs)**

GSM system architecture, Radio interface, Protocols, Localization and calling, Handover, Authentication and security in GSM, GSM speech coding, Concept of spread spectrum, Architecture of IS-95 CDMA system, Air interface, CDMA forward channels, CDMA reverse channels, Soft handoff, CDMA features, Power control in CDMA, Performance of CDMA System, RAKE Receiver, CDMA2000 cellular technology, GPRS system architecture.

UNIT V RECENT WIRELESS TECHNOLOGIES**(9 Hrs)**

Introduction to Wi-Fi, WiMAX, ZigBee Networks, Software Defined Radio, UWB Radio, Cognitive radio, Ad-hoc and sensor network, evolution of wireless network, MIMO systems, Security issues and challenges in a Wireless network.

Text Books

1. T.S. Rappaport, "Wireless Communication-Principles and practice", Pearson Publications, 2nd Edition, 2010.
2. Steve Rackley, Wireless Networking Technology, From Principles to Successful Implementation, Newnes; 1st edition, 2011
3. Gottapu Sasibhushana Rao, "Mobile Cellular Communication", Pearson Education, 2012

Reference Books

1. Upena Dalal and Manoj K. Shukla, "Wireless and Mobile Communication", Oxford Press Publications, 2016.
2. Andrea Goldsmith, "Wireless Communications", Cambridge University Press, 2012.
3. Ezio Biglieri and Robert Calderbank, "MIMO Wireless Communications", Cambridge University Press, 2015.
4. Kaveh Pah Laven and P. Krishna Murthy, "Principles of Wireless Networks", Pearson Education, 2012
5. William Stallings, "Wireless Communication and Networking", PHI, 2003.

Web References

1. <http://nptel.ac.in/courses/117102062/>
2. https://onlinecourses.nptel.ac.in/noc17_cs37/
3. <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-452-principles-of-wireless-communications-spring-2006/>
4. https://jiscollege.ac.in/ece/Syllabus_MCNT_2018.pdf
5. <https://learnengineering.in/ec8652-wireless-communication/>

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1	3	2	1	-	-	-	-	-	-	-	1	1	3	2	-
2	3	2	1	-	1	-	-	-	-	-	1	1	3	2	-
3	3	2	1	-	1	-	-	-	-	-	1	1	3	2	-
4	3	2	1	-	-	-	-	-	-	-	1	1	3	2	-
5	3	2	1	-	-	-	-	-	-	-	1	1	3	2	-

Correlation Level: 1-Low, 2-Medium, 3-High

U20CCT616

DATA SCIENCE

L	T	P	C	Hrs
3	0	0	3	45

Course Objectives

- To understand the concepts of Real world data representations.
- To learn the Data collection and its strategies in Data Science.
- To understand the Data analytics for Data Science.
- To gain knowledge in Data Science Tools.
- To expose the different opportunities in Industries.

Course outcomes

After completion of the course, the students will be able to

CO1 – Infer the Real world data and information. **(K2)**

CO2 – Explain basic concepts of Data collection and Data Preprocessing **(K2)**

CO3 – Make use of descriptive statistics to organize and summarize the data. **(K2)**

CO4 – Interpret the various Tools and its advantage. **(K3)**

CO5 – Illustrate the different opportunities in Industries. **(K2)**

UNIT I INTRODUCTION**(9 Hrs)**

Introduction to Data Science – Evolution of Data Science – Data Science Roles – Scope of Data Science - Stages in a Data Science Project – Data science process- Retrieving data – Cleansing, integrating and transforming data – Data analysis – Build the models – Presenting findings and building applications.– Data Security Issues.

UNIT II DATA COLLECTION AND DATA PRE-PROCESSING**(9 Hrs)**

Data manipulation: Reading and selection – Filtering missing data – Sorting – Grouping – Ranking and plotting; Introduction to Python; Fundamental Python Libraries for Data Scientists : Numpy – Scipy – ScikitLearn – Pandas – Matplotlib; IDE; Data Manipulation with Pandas; Sample programs to pre-process and visualize data

UNIT II EXPLORATORY DATA ANALYTICS**(9 Hrs)**

Introduction – Data Preparation – Exploratory Data Analysis: Data summarization – Data distribution – Outlier Treatment – Measuring asymmetry – Continuous distribution; Estimation: Mean – Variance – Skewness and Kurtosis - Box Plots - Pivot Table - Heat Map - Correlation Statistics - ANOVA. Sampling – Covariance – Correlation.

UNIT IV DATA SCIENCE TOOLS**(9 Hrs)**

Introduction – Frequentist Approach – Measuring the Variability in Estimates: Point estimates – Confidence intervals; Hypothesis Testing: Using confidence intervals – Using p-values. Introduction to Data Science Tool – Data Cleaning Tools – Data Munging and Modelling Tools -Visualization Tools

UNIT V APPLICATION**(9 Hrs)**

Data Economy and Industrialization – Introduction: Data Economy – Data Industry – Data Services – Data Science Application: Introduction – General Application Guidance – Different Domain – Advertising – Aerospace and Astronomy – Arts – Creative Design and Humanities – Bioinformatics – Consulting Services – Ecology and Environment – Ecommerce and Retail – Education – Engineering – Finance and Economy – Gaming.

Text Books

1. Chirag Shah, “A Hands on Introduction to Data Science”, Cambridge University Press, 2020.
2. SinanOzdemir, “Principles of Data Science”, Packt Publication, 2016.
3. Jiawei Han, Micheline Kamber and Jian Pei, “ Data Mining: Concepts and Techniques”, Third Edition. ISBN 0123814790, 2011.

Reference Books

1. Steven S. Skiena, "Data Science Design Manual", Springer International Publication, 2017.
2. Davy Cielen, Arno D B Meysman, Mohamed Ali, "Introducing Data Science – Big data, Machine Learning, and more using Python tools", Manning Publications Co, 2016.
3. Laura Igual, Santi Segua, "Introduction to Data Science – A Python Approach to Concepts, Techniques and Applications", Springer Nature, 2017
4. Cathy O'Neil and Rachel Schutt, "Doing Data Science", O'Reilly, 2015.
5. Mohammed J. Zaki and Wagner Miera Jr, "Data Mining and Analysis: Fundamental Concepts and Algorithms", Cambridge University Press, 2014.

Web References

1. <https://www.coursera.org/learn/excel-data-analysis>
2. https://www.tutorialspoint.com/excel_data_analysis/index.htm
3. <https://www.coursera.org/learn/open-source-tools-for-data-science>
4. <https://www.jeremyjordan.me/data-science>
5. <https://www.ngdata.com/top-data-science-resources>

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1	2	2	2	1	1	-	-	-	-	-	-	-	2	2	2
2	2	2	2	2	2	-	-	-	-	-	-	-	2	2	2
3	2	2	2	2	2	-	-	-	-	-	-	-	2	2	2
4	3	3	3	3	3	-	-	-	-	-	-	-	3	3	3
5	3	2	2	2	2	-	-	-	-	-	-	-	3	2	2

Correlation Level: 1-Low, 2-Medium, 3- High

U20CCP609	INTERNET OF THINGS APPLICATION LABORATORY	L	T	P	C	Hours
		0	0	2	1	45

Course Objectives

- To impart necessary and practical knowledge of components of Internet of Things.
- To attain the knowledge about different types of architecture and their elements of IoT.
- To understand the concepts of integration of devices and data's.
- To acquire the knowledge about remotely monitor data and control devices.
- To develop skills required to build real-time IoT based projects.

Course Outcomes

After completion of the course, students will be able to

CO1-Demonstrate internet of Things and its hardware and software components.

CO2- Interface I/O devices, sensors & communication modules.

CO3- Explain the concepts of remotely monitor data and control devices.

CO4- Build and deploy an various architecture with their elements.

CO5- Develop real life IoT based projects.

LIST OF EXPERIMENTS

1. Study Raspberry Pi and perform necessary software installation.
2. Introduction to Raspberry PI platform and python programming
3. Interface LED/Buzzer with Raspberry Pi
4. To interface Push button/Digital sensor (IR/LDR) with Raspberry Pi
5. Interface temperature and humidity sensor with Raspberry Pi
6. Interface DC motor using motor driver circuit with Raspberry Pi and control its rotation
7. Interface Bluetooth with Raspberry Pi and perform real time application
8. Write a program on Raspberry Pi to upload temperature and humidity data to thingspeak cloud and retrieve data from thingspeak cloud.
9. Communicate between Arduino and Raspberry PI using any wireless medium
10. To install MySQL database on Raspberry Pi and perform basic SQL queries.
11. Write a program on Raspberry Pi to publish and subscribe log data to MQTT broker.
12. To create TCP server on Raspberry Pi and respond with data to TCP client when requested.
13. To create UDP server on Raspberry Pi and respond with humidity data to UDP client when requested.
14. LoRaWAN Configuration to share the data in cloud.

Reference Books

1. Jeeva Jose, "Internet of Things", Khanna Publishing House, Delhi, 2012
2. Adrian McEwen, "Designing the Internet of Things", Wiley, 2007
3. Raj Kamal, "Internet of Things: Architecture and Design", McGraw Hill, 2002
4. CunoPfister, "Getting Started with the Internet of Things", O Reilly Media, 2015
5. Pethuru Raj and Anupama C. Raman, "The Internet of Things: Enabling Technologies, Platforms, and Use Cases", CRC Press, 2017

Web References

1. <https://www.i-scoop.eu/internet-of-things-guide/>
2. <https://www.theinternetofthings.eu/>
3. <https://www.udemy.com/course/complete-guide-to-build-iot-things-from-scratch-to-market/>
4. <https://www.coursera.org/learn/iot>

COs/POs/PSOs Mapping

COs	Program Outcomes (POs)												Program Specific Outcomes (PSOs)		
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
1	2	1	-	-	3	-	-	-	-	-	-	-	3	2	2
2	2	1	-	-	3	-	-	-	-	-	-	-	3	3	2
3	3	2	1	1	3	-	-	-	-	-	-	-	3	2	2
4	3	2	1	1	3	-	-	-	-	-	-	-	3	3	3
5	3	2	1	1	3	-	-	-	-	-	-	-	3	3	3

Correlation Level: 1-Low, 2-Medium, 3-High

U20CCP610	INTERNET PROGRAMMING LABORATORY	L	T	P	C	Hrs
		0	0	3	1	45

Course Objectives

- To be familiar with Web page design using HTML/XML and style sheets
- To be exposed to creation of user interfaces using Java frames and applets.
- To learn to create dynamic web pages using server side scripting.
- To be familiar with the PHP programming.
- To be exposed to creating applications with AJAX.

Course Outcomes

After completion of the course, the students will be able to

- CO1-** Construct Web pages using HTML/XML and style sheets. **(K3)**
CO2- Build dynamic web page with validation using Java Script objects and by applying different event handling mechanisms. **(K3)**
CO3- Develop dynamic web pages using server side scripting. **(K3)**
CO4- Use PHP programming to develop web applications. **(K3)**
CO5- Construct web applications using AJAX and web services. **(K3)**

LIST OF EXPERIMENTS

1. Create a web page with the following using HTML
 - a. To embed a map in a web page
 - b. To fix the hot spots in that map
 - c. Show all the related information when the hot spots are clicked.
2. Create a web page with the following.
 - a. Cascading style sheets.
 - b. Embedded style sheets.
 - c. Inline style sheets. Use our college information for the web pages.
3. Validate the Registration, user login, user profile and payment by credit card pages using JavaScript.
4. Write programs in Java using Servlets:
 - i. To invoke servlets from HTML forms
 - ii. Session tracking using hidden form fields and Session tracking for a hit count
5. Write programs in Java to create three-tier applications using servlets for conducting on-line examination for displaying student mark list. Assume that student information is available in a database which has been stored in a database server.
6. Install TOMCAT web server. Convert the static web pages of programs into dynamic web pages using servlets (or JSP) and cookies. Hint: Users information (user id, password, credit card number) would be stored in web.xml. Each user should have a separate Shopping Cart.
7. Redo the previous task using JSP by converting the static web pages into dynamic web pages. Create a database with user information and books information. The books catalogue should be dynamically loaded from the database.
8. Create and save an XML document at the server, which contains 10 users Information. Write a Program, which takes user Id as an input and returns the User details by taking the user information from the XML document
9. Write programs in PHP
 - i. Validate the form using PHP regular expression.
 - ii. PHP stores a form data into database.
10. Write a web service for finding what people think by asking 500 people's opinion for any consumer product.
11. Create an element that you can't change its children or attributes. An element is like a single frame in a movie: it represents the UI at a certain point in time using React elements.
12. Create a components in React Js that renders Welcome many times.

Reference Books

1. Deitel and Deitel and Nieto, "Internet and World Wide Web How to Program", Pearson Education, 5th Edition, 2011.
2. Robert W. Sebesta, "Programming with World Wide Web", Pearson Education, Eighth Edition, 2015.
3. Stephen Wynnkoop and John Burke, "Running a Perfect Website", QUE, 2nd Edition, 1999.
4. Chris Bates, Web Programming, "Building Intranet Applications", Wiley Publications, 3rd Edition, 2009.
5. Jeffrey C and Jackson, "Web Technologies A Computer Science Perspective", Pearson Education, 2011.

Web Reference

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3	3	3	2	1	2	-	-	-	-	-	1	1	2	3	2
4	3	3	2	1	2	-	-	-	-	-	1	1	2	3	2
5	3	3	2	1	2	-	-	-	-	-	1	1	2	3	2

Correlation Level: 1-Low, 2-Medium, 3- High

WIRELESS COMMUNICATION SYSTEMS LABORATORY		L	T	P	C	Hours
U20CCP611		0	0	3	1	45

Course Objectives

- To perform the various multiple access technique using trainer kit.
- To Learn the GPS receiver and demonstrate how to measure latitude & longitude.
- To understand and apply various Error correction codes.
- To understand about multipath receiver.
- To know the concept of various generation in wireless communication.

Course Outcomes

After completion of the course, the students are able to

CO1 - Analyze the various multiple access technique using trainer kit.

CO2- Demonstrate the transmission and reception of audio, video and tone using satellite communication trainer.

CO3 - Distinguish between different codes.

CO4 - Implement multipath receiver.

CO5 - Familiarize different generation in wireless communication.

LIST OF EXPERIMENTS

1. To perform Modulation, Demodulation and BER measurement using CDMA – DSSS Trainer.
2. To establish analog/digital communication link and transmit & receive three signals (audio, video, tone) simultaneously using Satellite Communication Trainer.
3. To study GPS Receiver, establishing link between GPS satellite & GPS trainer and measure of latitude & longitude
4. Simulation of error correction code (like CRC).
5. To study FHSS Transceiver
6. To study RAKE Receiver.
7. To study GSM Transceiver system.
8. To Study Bluetooth system kit.
9. To design and implement m-sequence generator
10. Simulation of Channel model for Free space propagation loss and log normal shadowing models
11. Simulation of Frequency Division Multiple access transmitter and receiver systems
12. BER simulation of OFDM system over multipath fading channel
13. Simulation of Frequency Division Multiple access techniques for communication systems
14. Simulation of CDMA transmitter and receiver
15. Simulation of Direct sequence spread spectrum modulation and demodulation
16. Generation of OFDM Transmitter and receiver systems using SDR kit
17. Design and Lead Software Defined Network experiments based both on Virtual Machines
18. Program Python scripts for SDN

Reference Books

1. Upena Dalal and Manoj K. Shukla, "Wireless and Mobile Communication", Oxford Press Publications, 2016.
2. Andrea Goldsmith, "Wireless Communications", Cambridge University Press, 2012.
3. Ezio Biglieri and Robert Calderbank, "MIMO Wireless Communications", Cambridge University Press, 2015.

4. Vijay K. Garg, "Wireless Communications and Networks", Morgan Kaufmann Publishers an Imprint of Elsevier, USA 2009 (Indian reprint).
5. J. Schiller, "Mobile Communication" 2/e, Pearson Education, 2012.

Web References

1. <http://nptel.ac.in/courses/117102062/>
2. https://onlinecourses.nptel.ac.in/noc17_cs37/
3. <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-452-principles-of-wireless-communications-spring-2006>
4. http://www.mnit.ac.in/dept_ece/download/Syllabus_Wireless_Optical.pdf
5. <https://www.abebooks.com/book-search/title/wireless-communication-technology/author/blake/>

COs / POs / PSOs Mapping

COs	Program Outcomes (POs)												Program Specific Outcomes (PSOs)		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	1	1	1	2	-	-	-	2	-	-	2	3	-	2
2	3	1	1	1	2	-	-	-	2	-	-	2	3	-	2
3	3	1	1	1	2	-	-	-	2	-	-	2	3	-	2
4	3	1	1	1	2	-	-	-	2	-	-	2	3	-	2
5	3	1	1	1	2	-	-	-	2	-	-	2	3	-	2

Correlation Level: 1-Low, 2-Medium, 3-High

U20CCC6XX	CERTIFICATION COURSE - VI	L	T	P	C	Hours
		0	0	4	-	45

Students shall choose an International certification course offered by the reputed organizations like Google, Microsoft, IBM, Texas Instruments, Bentley, Autodesk, Eplan and CISCO, etc. The duration of the course is 40-50 hours specified in the curriculum, which will be offered through Centre of Excellence.

Pass /Fail will be determined on the basis of participation, attendance, performance and completion of the course. If a candidate fails, he/she has to repeat the course in the subsequent years. Pass in this course is mandatory for the award of degree.

U20CCS606	SKILL DEVELOPMENT COURSE 6 (Foreign Language / IELTS – II)	L	T	P	C	Hours
		0	0	2	-	30

Student should choose the Foreign Language/IELTS course like Japanese/French/ Germany/IELTS, etc. approved by the Department committee comprising of HoD, Programme Academic Coordinator, Class advisor and language Experts. The courses are to be approved by Academic Council on the recommendation of HoD at the beginning of the semester if necessary, subject to ratification in the next Academic council meeting. Students have to complete the courses successfully. The Committee will monitor the progress of the student and recommend the grade (100% Continuous Assessment pattern) based on the completion of course. The marks attained for this course is not considered for CGPA calculation

U20CCS607	SKILL DEVELOPMENT COURSE 7 (Technical Seminar)	L 0	T 0	P 2	C -	Hours 30
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Course Objectives

- To encourage the students to study advanced engineering developments
- To prepare and present technical reports.
- To encourage the students to use various teaching aids such as over head projectors, power point presentation and demonstrative models.

Course Outcomes

After completion of the course, the students will be able to

CO1 - Review, prepare and present technological developments.

CO2 - Face the placement interviews.

Method of Evaluation:

- During the seminar session each student is expected to prepare and present a topic on engineering/technology, for duration of about 20 minutes.
- In a session of three periods per week, 8 to 10 students are expected to present the seminar.
- Each student is expected to present atleast twice during the semester and the student is evaluated based on that.
- At the end of the semester, he / she can submit a report on his / her topic of seminar and marks are given based on the report.
- A Faculty guide is to be allotted and he / she will guide and monitor the progress of the student and maintain attendance also.
- Evaluation is 100% internal. The marks attained for this course is not considered for CGPA calculation.

U20CCS608	SKILL DEVELOPMENT COURSE 8 (NPTEL / MOOC - I)	L	T	P	C	Hrs
		0	0	0	-	50

Student should register online courses like MOOC / SWAYAM / NPTEL etc. approved by the Department committee comprising of HoD, Programme Academic Coordinator, Class advisor and Subject Experts. Students have to complete the relevant online courses successfully. The list of online courses is to be approved by Academic Council on the recommendation of HoD at the beginning of the semester if necessary, subject to ratification in the next Academic council meeting. The Committee will monitor the progress of the student and recommend the grade (100% Continuous Assessment pattern) based on the completion of course / marks secured in online examinations. The marks attained for this course is not considered for CGPA calculation.

U20CCM606	ESSENCE OF INDIAN TRADITIONAL KNOWLEDGE	L 0	T 0	P 2	C -	Hours 30
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Course Objectives

- To get a knowledge in Indian Culture
- To Know Indian Languages and Literature and the fine arts in India
- To explore the Science and Scientists of Medieval and Modern India

Course Outcomes

After completion of the course, the students will be able to

- CO1- Understand philosophy of Indian culture.
 CO2 -Distinguish the Indian languages and literature.
 CO3 -Learn the philosophy of ancient, medieval and modern India.
 CO4 - Acquire the information about the fine arts in India.
 CO5 - Know the contribution of scientists of different eras.

UNIT - I Introduction to Culture:

Culture, civilization, culture and heritage, general characteristics of culture, importance of culture in human literature, Indian Culture, Ancient India, Medieval India, Modern India

UNIT - II Indian Languages, Culture and Literature:

Indian Languages and Literature-I: the role of Sanskrit, significance of scriptures to current society, Indian philosophies, other Sanskrit literature, literature of south India Indian Languages and Literature-II: Northern Indian languages & literature

UNIT - III Religion and Philosophy:

Religion and Philosophy in ancient India, Religion and Philosophy in Medieval India, Religious Reform Movements in Modern India (selected movements only)

UNIT – IV Fine Arts in India (Art, Technology& Engineering):

Indian Painting, Indian handicrafts, Music, divisions of Indian classic music, modern Indian music, Dance and Drama, Indian Architecture (ancient, medieval and modern), Science and Technology in India, development of science in ancient, medieval and modern India

UNIT – V Education System in India:

Education in ancient, medieval and modern India, aims of education, subjects, languages, Science and Scientists of Ancient India, Science and Scientists of Medieval India, Scientists of Modern India

Reference Books

1. Kapil Kapoor, "Text and Interpretation: The India Tradition", ISBN: 81246033375, 2005
2. "Science in Samskrit", Samskrita Bharti Publisher, ISBN 13: 978-8187276333, 2007
3. NCERT, "Position paper on Arts, Music, Dance and Theatre", ISBN 81-7450 494-X, 200
4. S. Narain, "Examinations in ancient India", Arya Book Depot, 1993
5. Satya Prakash, "Founders of Sciences in Ancient India", Vijay Kumar Publisher, 1989
6. M. Hiriyanna, "Essentials of Indian Philosophy", Motilal Banarsidass Publishers, ISBN 13: 978-8120810990, 2014

PROFESSIONAL ELECTIVE-III

	DIGITAL IMAGE PROCESSING	L	T	P	C	Hrs
U20ECCM03	(Common to ECE and CCE)	3	0	0	3	45

Course Objectives

- To become familiar with digital image fundamentals
- To get exposed to mathematical preliminaries and Image Transform used in Image Processing
- To learn concepts of Image Enhancement and restoration techniques.
- To study the image segmentation and colour Image processing techniques.
- To become familiar with image compression and recognition methods

Course Outcomes

After completion of the course, the students will be able to

CO1- Summarize the digital image fundamentals. **(K2)**

CO2- Correlate the various image processing technique with the help of mathematical preliminaries. **(K4)**

CO3- Apply different types of image enhancement and restoration techniques in various applications **(K3)**

CO4- Illustrate the significance of Colour Image Processing and Image Segmentation techniques **(K4)**

CO5- Connect Image compression and Recognition techniques in Image processing. **(K4)**

UNIT I DIGITAL IMAGE FUNDAMENTALS**(9 Hrs)**

Introduction – Origin – Steps in Digital Image Processing – Components – Elements of Visual Perception – Image Sensing and Acquisition – Image Sampling and Quantization – Relationships between pixels., simple image formation model, Brightness, contrast, hue, saturation, Mach band effect

UNIT II IMAGE TRANSFORM:**(9 Hrs)**

Two dimensional Fourier Transform- Properties – Fast Fourier Transform – Inverse FFT- Image transforms – 1D DFT, 2D DFT, Discrete Cosine transform, Discrete Sine transform, Hadamard transform, Haar transform, Slant transform, KL transform, SVD transform, Wavelet transform.

UNIT III IMAGE ENHANCEMENT AND IMAGE RESTORATION**(9 Hrs)**

Spatial Domain: Gray level transformations – Histogram processing – Basics of Spatial Filtering – Smoothing and Sharpening Spatial Filtering – Frequency Domain: Introduction to Fourier Transform – Smoothing and Sharpening frequency domain filters – Ideal, Butterworth and Gaussian filters. Noise models – Mean Filters – Order Statistics – Adaptive filters – Band reject Filters – Band pass Filters – Notch Filters – Optimum Notch Filtering – Inverse Filtering – Wiener filtering.

UNIT IV COLOUR IMAGE PROCESSING AND IMAGE SEGMENTATION**(9 Hrs)**

Colour fundamentals – Colour models – HIS to RGB and RGB to HIS. Detection of Discontinuities – Edge Linking and Boundary detection – Region based segmentation- Morphological processing- erosion and dilation. Segmentation by morphological watersheds – basic concepts – Dam construction – Watershed segmentation algorithm

UNIT V IMAGE COMPRESSION AND RECOGNITION**(9 Hrs)**

Need for compression – Coding Redundancy - Interpixel Redundancy - Psycho visual Redundancy - Bit plane coding - Variable length coding – Adaptive coding – Arithmetic coding – LZW coding – Hybrid coding – Wavelet – JPEG – MPEG.. Boundary representation, Boundary description, Fourier Descriptor, Regional Descriptors – Topological feature, Texture – Patterns and Pattern classes – Recognition based on matching.

Text Books

1. Rafael C. Gonzalez & Richard E. Woods, Digital Image Processing, 2017, 4th edition, Pearson Education, USA
2. Anil K. Jain, Fundamentals of Digital Image Processing, 2015, 1st edition, Pearson India, India
3. Kenneth R. Castleman, Digital Image Processing, Pearson, 2006.

Reference Books

1. Rafael C. Gonzalez, Richard E. Woods, Steven L. Eddins, "Digital Image Processing Using MATLAB", Third Edition Tata Mc Graw Hill Pvt. Ltd., 2011.
2. William K Pratt, "Digital Image Processing", John Wiley, 2002.
3. Malay K. Pakhira, "Digital Image Processing and Pattern Recognition", First Edition, PHI Learning Pvt. Ltd., 2011.
4. John C. Russ, F. Brent Neal-The Image Processing Handbook, Seventh Edition, The Kindle edition (2016), CRC Press, Taylor & Francis Group.
5. P.Ramesh Babu, Digital Image Processing, Scitech Publications., 2003

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1. <http://eeweb.poly.edu/~onur/lectures/lectures.html>
2. <http://www.caen.uiowa.edu/~dip/LECTURE/lecture.html>
3. <https://nptel.ac.in/courses/117/105/117105079/>
4. <https://nptel.ac.in/courses/117/105/117105135/>
5. <https://www.csie.nuk.edu.tw/>

COs/POs/PSOs Mapping

COs	Program Outcomes (POs)												Program Specific Outcomes (PSOs)		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	2	2	2	-	-	-	-	-	-	-	-	2	1	3
2	3	2	2	2	-	-	-	-	-	-	-	-	2	1	3
3	3	2	2	2	-	-	-	-	-	-	-	-	2	1	3
4	3	2	2	2	-	-	-	-	-	-	-	-	2	1	3
5	3	2	2	2	-	-	-	-	-	-	-	-	2	1	3

Correlation Level: 1-Low, 2-Medium, 3- High

WIRELESS MOBILE NETWORKING 3 0 0 3 45**Course Objectives**

- To understand the functioning of wireless communication system and evolution of different wireless communication systems and standards.
- To explain GPRS and multiple access techniques for Wireless Communication
- To explain wireless application on protocol and features of mobile services
- To understand the key modules of digital communication systems and encoding techniques
- To evaluate design challenges, constraints and security issues associated with Ad-hoc wireless networks

Course Outcomes

After completion of the course, the students will be able to

- CO1-** Describe the function of PCS and call processing in GSM
CO2- Demonstrate an ability explain GPRS and multiple access techniques for Wireless Communication
CO3- Compare the features of next generation mobile communication networks.
CO4- Describe and analyze the digital communication system with spread spectrum modulation.
CO5- Demonstrate an ability to evaluate design challenges, constraints and security issues associated with Ad-hoc wireless networks.

UNIT I BASICS OF PCS AND GSM**(9 Hrs)**

Personal Communication Services (PCS) architecture- Global system for Mobile Communication (GSM) Architecture, GSM frequency spectrum, GSM radio aspects, GSM services, Supplementary services, GSM channel types, call processing in GSM- Mobility Management: Location updates procedure, Temporary Mobile Subscriber Identity, concept of roaming, Location area, routing area, tracking area

UNIT II GPRS AND MOBILE DATA COMMUNICATION**(9 Hrs)**

General Packet Radio Services (GPRS) architecture, GPRS Services, Quality of service – GPRS Network nodes, Mobility management and routing in GPRS- WLANs (wireless LANs) IEEE 802.11 standard, RFID- Bluetooth technology, Wi-Max, Wi-Fi- Mobile IP.

UNIT III WIRELESS APPLICATION ON PROTOCOL AND 3G MOBILE SERVICE (9 Hrs)

Mobile Internet standard, WAP Gateway and Protocols- WML- IMT 2000 specification- W-CDMA, CDMA 2000, Quality of services in 3G Network- UMTS Technology- Features of 4G, 4G LTE, VoLTE, 4.5G, 5G.

UNIT IV WLL, SIGNAL ENCODING TECHNIQUES AND SS MODULATION**(9 Hrs)**

WLL architecture and technologies- WLL types and application- Concept of LEC networks- Line coding techniques- ASK, BPSK, PCM, DPCM and DM- Types of spread spectrum: DSSS, FHSS

UNIT V APPMOBILE AD-HOC NETWORK AND WSN**(9Hrs)**

MANET, MANET topologies, Features of MANET, Applications, types of MANET Architecture, Design challenges in MANET- Mesh Networking- Wireless sensor network, Applications, Clustering of WSN, Characteristics of WSN- Sensor node: Block diagram, different types of WSN Architecture, Energy efficiency in WSN.

Text Books

1. Theodore S. Rappaport, "Wireless communication- Principles and practice", Pearson publication New Delhi, 2005
2. Lin Yi-Bang, Clamta Imrich, "Wireless and mobile network Architectures", John Wiley & sons, New Delhi, 2001
3. T.G.Palanivelu, "Wireless and Mobile Communication", PHI learning pvt ltd, 2008

Reference Books

1. Singal T.L., "Wireless Communication", McGraw Hill Education Private Limited, New Delhi, 2010,
2. Talukdar Asoke K, Javagal Roopa R, "Mobile Computing Technology", Hill Education Private Limited, New Delhi, 2010,
3. Willam Stallings, "Wireless Communications and Networks", 2nd edition, Pearson Publisher, 2004
4. Simon Haykin · Michael Moher, "Modern Wireless Communications, Pearson Prentice Hall, 2005
5. Iti Saha Misra, "Wireless Communications and Networks: 3G and Beyond", McGraw Hill Education (India) Pvt Ltd, 2013

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1. https://www.octoscope.com/English/Collaterals/Presentations/octoScope_WirelessTutorial_20090209.pdf
2. www.radio-Electronics.com/info/wireless/Bluetooth/Bluetooth_overview.php
3. <https://www.data-alliance.net/blog/5g-4g-3g-standards-lte-gsm-cdma-ism-wcdma-hspa>

COs/POs/PSOs Mapping

COs	Program Outcomes (POs)												Program Specific Outcomes (PSOs)		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	3	3	1	-	-	-	-	-	-	1	1	3	2	-
2	2	2	3	1	-	-	-	-	-	-	1	1	3	2	-
3	3	1	3	1	-	-	-	-	-	-	1	1	3	3	-
4	3	2	3	1	-	-	-	-	-	-	1	1	3	3	1
5	3	1	3	1	-	-	-	-	-	-	1	1	3	2	1

Correlation Level: 1-Low, 2-Medium, 3- High

U20CCE613	INFORMATION RETRIEVAL	L	T	P	C	Hrs
		3	0	0	3	45

Course Objectives

- To understand the basics of Information Retrieval.
- To learn retrieval evaluation model
- To understand machine learning techniques for text classification and clustering.
- To understand various search engine system operations.
- To learn different techniques of recommender system.

Course Outcomes

CO1- Basics of information retrieval models. **(K2)**

CO2- Use an open source search engine framework and explore its capabilities. **(K3)**

CO3- Apply appropriate method of classification or clustering. **(K3)**

CO4- Design and implement innovative features in a search engine. **(K3)**

CO5- Design and implement a recommender system. **(K3)**

UNIT I INTRODUCTION**(9 Hrs)**

Information Retrieval – Early Developments – The IR Problem – The Users Task – Information versus Data Retrieval – The IR System – The Software Architecture of the IR System – The Retrieval and Ranking Processes – The Web – The e-Publishing Era – How the web changed Search – Practical Issues on the Web – How People Search – Search Interfaces Today – Visualization in Search Interfaces.

UNIT II MODELING AND RETRIEVAL EVALUATION**(9 Hrs)**

Basic IR Models – Boolean Model – TF-IDF (Term Frequency/Inverse Document Frequency) Weighting – Vector Model – Probabilistic Model – Latent Semantic Indexing Model – Neural Network Model – Retrieval Evaluation – Retrieval Metrics – Precision and Recall – Reference Collection – User-based Evaluation – Relevance Feedback and Query Expansion – Explicit Relevance Feedback.

UNIT III TEXT CLASSIFICATION AND CLUSTERING**(9 Hrs)**

A Characterization of Text Classification – Unsupervised Algorithms: Clustering – Naïve Text Classification – Supervised Algorithms – Decision Tree – k-NN Classifier – SVM Classifier – Feature Selection or Dimensionality Reduction – Evaluation metrics – Accuracy and Error – Organizing the classes – Indexing and Searching – Inverted Indexes – Sequential Searching – Multi-dimensional Indexing.

UNIT IV WEB RETRIEVAL AND WEB CRAWLING**(9 Hrs)**

The Web – Search Engine Architectures – Cluster based Architecture – Distributed Architectures – Search Engine Ranking – Link based Ranking – Simple Ranking Functions – Learning to Rank – Evaluations – Search Engine Ranking – Search Engine User Interaction – Browsing – Applications of a Web Crawler – Taxonomy – Architecture and Implementation – Scheduling Algorithms – Evaluation.

UNIT V RECOMMENDER SYSTEM**(9 Hrs)**

Recommender Systems Functions – Data and Knowledge Sources – Recommendation Techniques – Basics of Content-based Recommender Systems – High Level Architecture – Advantages and Drawbacks of Content-based Filtering – Collaborative Filtering – Matrix factorization models – Neighborhood models.

Text Books

1. Ricardo Baeza-Yates and Berthier Ribeiro-Neto, "Modern Information Retrieval: The Concepts and Technology behind Search", ACM Press Books, Second Edition, 2011.
2. Ricci, F, Rokach, L. Shapira, B.Kantor, "Recommender Systems Handbook", First Edition, 2011.
3. W.Bruce Croft, Donald Metzler and Trevor Strohman, "Search Engines: Information Retrieval in Practice", Pearson, 1 st Edition, 2009.

Reference Books

1. C. Manning, P. Raghavan, and H. Schütze, "Introduction to Information Retrieval", Cambridge University Press, 2008.
2. Stefan Buettcher, Charles L. A. Clarke and Gordon V. Cormack, "Information Retrieval: Implementing and Evaluating Search Engines", The MIT Press, 2010.
3. Gerald J. Kowalski, Mark T. Maybury, "Information Storage and Retrieval Systems: Theory and Implementation", Kluwer Academic Publishers, second edition, 2000.
4. Ricardo Baeza-Yates, "Modern Information Retrieval", Pearson Education, 2007.
5. Ophir Frieder "Information Retrieval: Algorithms and Heuristics: The Information Retrieval Series ", Springer, 2 nd Edition, 2004.

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1. <https://tinyurl.com/ydf33ye6>
2. <https://nptel.ac.in/courses/106/105/106105081/>
3. <https://nptel.ac.in/courses/106/105/106105183/>

COs/POs/PSOs Mapping

COs	Program Outcomes (POs)												Program Specific Outcomes (PSOs)		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	1	2	1	-	-	-	-	-	1	-	1	2	3	-
2	3	1	2	1	-	-	-	-	-	1	-	1	2	3	-
3	3	1	2	2	-	-	-	-	1	1	1	1	2	3	-
4	3	1	2	1	-	-	-	-	1	1	1	1	2	3	-
5	3	1	2	2	-	-	-	-	1	1	1	1	2	3	-

Correlation Level: 1-Low, 2-Medium, 3- High

U20CSCM05	HUMAN COMPUTER INTERACTION	L	T	P	C	Hrs
	(Common to CSE and CCE)	3	0	0	3	45

Course Objectives

- To Learn the foundations of Human Computer Interaction
- To be familiar with the design technologies for individuals and persons with disabilities
- To demonstrate the communication using multimedia and www.
- To study about mobile ecosystem.
- To learn about user interface.

Course Outcomes

After completion of the course, the students will be able to

CO1 - Collect fundamentals of human interaction and problem solving (**K2**)

CO2 - Design effective HCI for individuals..(**K3**)

CO3 - Enumerate the cognitive computerized models for HCI(**K2**)

CO4 - Design mobile application framework using HCI tools.(**K3**)

CO5 - Develop web interface using various tools(**K4**)

UNIT I INTRODUCTION**(9 Hrs)**

The Human: I/O channels – Memory – Reasoning and problem solving; The computer: Devices – Memory – processing and networks; Interaction: Models – frameworks – Ergonomics– styles – elements – interactivity- Paradigms.

UNIT II INTERACTIVE DESIGN AND RULE**(9 Hrs)**

Interactive Design basics – process – scenarios – navigation – screen design – Iteration and prototyping. HCI in software process – software life cycle – usability engineering Prototyping in practice – design rationale. Design rules – principles, standards, guidelines, rules. Evaluation Techniques – Universal Design.

UNIT III COGNITIVE COMPUTERIZED MODELS**(9 Hrs)**

Cognitive models –Socio-Organizational issues and stake holder requirements –Communication and collaboration models-Hypertext, Multimedia and WWW.

UNIT IV APPLICATION FRAMEWORK**(9 Hrs)**

Mobile Ecosystem: Platforms, Application frameworks- Types of Mobile Applications: Widgets, Applications, Games- Mobile Information Architecture, Mobile 2.0, Mobile Design: Elements of Mobile Design, Tools.

UNIT V DESIGNING WEB INTERFACES**(9 Hrs)**

Designing Web Interfaces – Drag & Drop, Direct Selection, Contextual Tools, Overlays, Inlays and Virtual Pages, Process Flow. Case Studies.

Text Books

1. Interaction Design: Beyond Human-Computer Interaction by Helen Sharp & Jenny Preece & Yvonne Rogers 2019.
2. Interaction Design - Beyond Human-Computer Interaction, 4th Edition, 2016
3. Computer-Human Interaction. Cognitive Effects of Spatial Interaction, Learning, and Ability by Theodor Wyeld & Paul Calder & Haifeng Shen (eds.) ,2015
4. Designing Interactive Systems: A Comprehensive Guide to HCI, UX and Interaction Design. by David R. Benyon 2013.
5. Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale, "Human Computer Interaction", 3rd Edition, Pearson Education, 2004.

Reference Books

1. Brian Fling, "Mobile Design and Development", Fourth Edition, O'Reilly Media Inc., 2018.
2. Cabrera & James, "Modular Design Frameworks: A Projects-based Guide for UI/UX Designers", 2017.
3. Bill Scott and Theresa Neil, "Designing Web Interfaces", First Edition, O'Reilly 2009.
4. Jonathan Lazar, "Research Methods in Human-Computer Interaction", 2009.
5. Yvonne Rogers, "Interaction Design: Beyond Human-Computer Interaction", 2001.

Web References

1. <https://www.interaction-design.org/literature/topics/human-computer-interaction>
2. <https://www.hcii.cmu.edu/academics/mhci>
3. <https://www.rit.edu/study/human-computer-interaction-ms>
4. <https://www.udacity.com/course/human-computer-interaction--ud400>
5. <https://xd.adobe.com/ideas/principles/human-computer-interaction/>

COs/POs/PSOs Mapping

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	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	3	3	3	3	3	3	3	-	2	3	-	3	-	3
2	3	3	3	3	-	3	-	3	-	2	-	2	2	-	-
3	2	2	2	2	2	2	3	3	-	3	3	-	2	-	2
4	2	2	2	2	2	2	-	3	-	3	-	3	3	-	-
5	3	3	3	3	3	3	3	3	-	3	3	3	3	-	3

Correlation Level: 1-Low, 2-Medium, 3- High

	L	T	P	C	Hrs
U20BMCM01					
SOFT COMPUTING					
(Common to ECE, ICE, BME, CCE)	3	0	0	3	45

Course Objectives

- To gain knowledge in the basics of fuzzy set.
- To learn the concepts of Optimization algorithms.
- To get familiarized with the learning schemes of neural networks.
- To understand with the design of fuzzy interface systems.
- To get exposed to neuro-fuzzy hybrid systems and its applications

Course Outcomes

After completion of the course the students will be able to

- CO1** - Gain knowledge in the basics of fuzzy set. **(K2)**
CO2 - Understand the concepts of optimization algorithms. **(K2)**
CO3 - Explain the terminologies and various learning schemes of Neural Networks. **(K3)**
CO4 - Gain knowledge in fuzzy interface systems. **(K3)**
CO5 - Familiarize with the neuro-fuzzy hybrid systems and its applications. **(K3)**

UNIT- I Introduction to Fuzzy Set (9 Hrs)

Fuzzy Sets – Basic Definition and Terminology – Set-theoretic Operations – Member Function Formulation and Parameterization – Fuzzy Rules and Fuzzy Reasoning – Extension Principle and Fuzzy Relations – Fuzzy If-Then Rules – Fuzzy Reasoning – Fuzzy Inference Systems – Mamdani Fuzzy Models – Sugeno Fuzzy Models – Input Space Partitioning and Fuzzy Modeling.

UNIT- II Optimization (9 Hrs)

Optimization - Derivative-based Optimization – Descent Methods – The Method of Steepest Descent – Classical Newton's Method – Step Size Determination – Derivative-free Optimization – Genetic Algorithms – Simulated Annealing – Random Search – Downhill Simplex Search.

UNIT- III Neural Networks (9 Hrs)

Introduction to Neural Networks - Supervised Learning Neural Networks – Perceptrons - Adaline – Back propagation Multilayer Perceptrons – Radial Basis Function Networks – Unsupervised Learning Neural Networks – Competitive Learning Networks – Kohonen Self-Organizing Networks – Learning Vector Quantization – Hebbian Learning.

UNIT- IV Fuzzy Interface Systems (9 Hrs)

Type of fuzzy interface-Adaptive networks based Fuzzy interface systems – structure and parameters of a fuzzy system-Classification and Regression Trees - Rule based structure identification - Neuro-Fuzzy controls - Simulated annealing – Evolutionary computation.

UNIT- V Applications of Hybrid Systems (9 Hrs)

Hybrid Systems -Neural Networks, Fuzzy Logic - LR-Type Fuzzy Numbers - Fuzzy Neuron - Fuzzy BP Architecture - Learning in Fuzzy BP- Inference by Fuzzy BP - Fuzzy Art Map: A Brief Introduction - Soft Computing Tools - Fuzzy Logic Controller.

Text Books

1. Rajasekaran, G.A.Vijayalakshmi Pai, "Neural Networks, Fuzzy Logic and Genetic Algorithm, Synthesis and Applications ", PHI Learning Pvt.Ltd. 2017.
2. S.N. Sivanandam and S.N. Deepa, "Principles of Soft Computing", Wiley Publications, 2nd Edition, 2011
3. Ross T. J., "Fuzzy Logic with Engineering Applications", McGraw Hill, 2016.

Reference Books

1. J.S.R.Jang, C.T.Sun and E.Mizutani, "Neuro-Fuzzy and Soft Computing", Pearson Education 2004
2. Timothy J.Ross, "Fuzzy Logic with Engineering Applications", McGraw- Hill, 2011.

3. Jyh-Shing Roger Jang, Chuen-Tsai Sun, Eiji Mizutani, "Neuro-Fuzzy and Soft Computing", Prentice-Hall of India, 2002.
4. Martin T. Hagan Howard B. Deruth and Mark Beale, "Neural Network Design", Thompson Learning, 2002.
5. James A. Freeman and David M. Skapura, "Neural Networks Algorithms, Applications, and Programming Techniques", Addison Wesley, 2003.

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2. <https://www.elprocus.com/soft-computing/>
3. <https://nptel.ac.in/courses/108/104/108104049/>
4. http://www.myreaders.info/html/soft_computing.html
5. <https://nptel.ac.in/courses/106/105/106105173/>

COs/POs/PSOs Mapping

COs	Program Outcomes (POs)												Program Specific Outcomes (PSOs)		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	1	1	1	1	1	-	-	-	-	-	1	3	2	-
2	3	1	1	1	1	1	-	-	-	-	-	1	3	2	-
3	3	1	1	1	1	1	-	-	-	-	-	1	3	2	-
4	3	2	1	2	2	1	-	-	-	-	-	1	3	2	-
5	3	2	1	2	2	1	-	-	-	-	-	2	3	2	-

Correlation Level: 1-Low, 2-Medium, 3- High

OPEN ELECTIVE-III

U20EEO603	CONVENTIONAL AND NON- CONVENTIONAL ENERGY SOURCES (Common to ECE, ICE, MECH, CIVIL, BME, Mechatronics, CCE, AI&DS, FT)	L	T	P	C	Hrs
		3	0	0	3	45

Course Objectives

- To get knowledge on the status of conventional and non-conventional energy resources in world.
- To have a clear idea about the operation of conventional power plant and its associated equipment's.
- To learn about the concept of energy harvesting of solar through thermal and PV module
- To understand the technological basis for harnessing wind energy.
- To get a clear knowledge on power generation using Ocean, Tidal Energy and Bio-Energy

Course Outcomes

After completion of the course, the students will be able to

CO1 – Identify the world and Indian energy scenario and the necessity of renewable energy sources **(K1)**

CO2 – Gain knowledge for the generation of electrical power from various power plants **(K1)**

CO3 – Analyze and compare the various solar harvesting techniques **(K3)**

CO4 – Describe the aerodynamics of wind turbines and calculate their power, energy production **(K1)**

CO5 – Describe the construction and working principle of various equipment's used in Ocean, Tidal Energy and Bio-Energy power plants **(K2)**

UNIT I ENERGY RESOURCES**(9 Hrs)**

Perspective of energy resources – Forms of Energy – Conventional and non-conventional sources of energy– World's energy status - Energy reserves in India. Limitations of Conventional sources of energy efficiency – Renewable Energy Sources – Energy parameters – Energy Intensity - Gross Domestic product.

UNIT II POWER PLANTS**(9 Hrs)**

Thermal power plant – layout, working principle. Gas turbine power plant – layout, working principle. Nuclear power plants: fuels, nuclear fuel cycle, reactors and nuclear waste management. Hydro Electric plants – Types, energy conversion schemes, environmental aspects.

UNIT III SOLAR ENERGY SYSTEMS**(9 Hrs)**

Solar radiation - Principles of solar energy collection –Types of collector – working principles - Characteristics - efficiency - Solar Energy applications – water heaters, air heaters, solar cooling; solar drying and power generation – solar tower concept – solar pump. Photovoltaic (PV) technology – photovoltaic effect – modelling -Characteristics – efficiency of solar cells.

UNIT IV WIND ENERGY SYSTEMS**(9 Hrs)**

General theory of wind mills – Types of wind mills – performance of wind machines–wind power – efficiency. Merits and Limitations of Wind energy system – Modes of wind power generation.

UNIT V ALTERNATE ENERGY SYSTEMS**(9 Hrs)**

Ocean and Tidal energy conversion - working principle of OTEC – Anderson closed cycle OTEC System. Tidal power – tides - tidal range - types of tidal power plants, single basin and double basins schemes. Bio-mass Energy – Biogas plants.

Text Books

1. S. Rao and Dr. B. B. Parulekar, "Energy Technology", Khanna Publication, 3rd Edition, 1999.
2. B. H. Khan, "Non-Conventional Energy Resources", Tata McGraw Hill Education, 2nd Edition, 2009.
3. D. P. Kothari, K. C. Singal, Rakesh Ranjan, "Renewable Energy Sources and Emerging Technologies", PHI, 2011

Reference Books

1. G. D. Rai, "Non-conventional energy sources", Khanna Publication. 4th Edition, 2002.
2. Pulfrey, David. L, "Photo voltaic Power Generation", Van Nostrand reinhold Company, 1983.
3. Abbasik, "Renewable Energy Sources and their Environment", PHI, 2008.
4. Steve Doty, Wayne C. Turner, "Energy Management Handbook", Fairmont Press, 8th Edition, 2012.
5. S.A.Abbasi and N. Abbasi, "Renewable Energy Sources and Their Environmental Impact", PHI, 2001.

Web References

1. https://www.tutorialspoint.com/renewable_energy/index.htm
2. <https://nptel.ac.in/courses/112/107/112107291/>
3. <https://byjus.com/physics/conventional-and-nonconventional-sources-of-energy/>
4. <https://www.jagranjosh.com/general-knowledge/nonconventional-sources-of-energy-1448698715-1>
5. <https://wb.gov.in/departments-power-and-non-conventional-energy-sources.aspx>

COs / POs and PSOs Mapping

COs	Program Outcomes (POs)												Program Specific Outcomes (PSOs)		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	1	1	2	-	1	2	-	-	-	-	1	-	1	1
2	3	1	1	2	-	1	2	-	-	-	-	1	-	1	1
3	3	1	1	2	-	1	2	-	-	-	-	1	-	1	1
4	3	1	1	2	-	1	2	-	-	-	-	1	-	1	1
5	3	1	1	2	-	1	2	-	-	-	-	1	-	1	1

Correlation Level: 1 - Low, 2 - Medium, 3 - High

U20EEO603	ELECTRONIC PRODUCT DESIGN AND PACKAGING					L	T	P	C	Hrs
	(Common to EEE, CSE, IT, ICE, MECH, CCE, BME, Mechatronics)					3	0	0	3	45

Course Objectives

- To provide basic knowledge about Electronic Product and Packaging
- To introduce and discuss various issues related to the system packaging
- To get clear idea about design of packages which can withstand higher temperature, vibrations and shock
- To Design of PCBs which minimize the EMI and operate at higher frequency
- To acquire depth knowledge about the concepts of Testing and testing methods

Course Outcomes

After completion of the course, students are able to

- CO1** - Explain the basics of Electronic Product and Packaging. **(K2)**
CO2 - Infer various issues related to the system packaging. **(K2)**
CO3 - Summarize the clear idea about design of packages which can withstand higher temperature, vibrations and shock **(K2)**
CO4 - Describe the design of PCBs which minimize the EMI and operate at higher frequency **(K2)**
CO5 - Explain the various testing methods **(K2)**

UNIT I : OVERVIEW OF ELECTRONIC SYSTEMS PACKAGING **(9 Hrs)**

Definition of a system and history of semiconductors, Products and levels of packaging, Packaging aspects of handheld products, Definition of PWB, Basics of Semiconductor and Process flowchart, Wafer fabrication, inspection and testing, Wafer packaging; Packaging evolution; Chip connection choices, Wire bonding, TAB and flip chip.

UNIT II : SEMICONDUCTOR PACKAGES **(9 Hrs)**

Single chip packages or modules (SCM), Commonly used packages and advanced packages; Materials in packages; Thermal mismatch in packages; Multichip modules (MCM)-types; System-in-package (SIP); Packaging roadmaps; Hybrid circuits;

UNIT III ELECTRICAL ISSUES IN PACKAGING **(9 Hrs)**

Electrical Issues of Systems Packaging, Signal Distribution, Power Distribution, Electromagnetic Interference, Transmission Lines, Clock Distribution, Noise Sources, Digital and RF Issues. Design Process Electrical Design: Interconnect Capacitance, Resistance and Inductance fundamentals; Packaging roadmaps – Hybrid circuits – Resistive, Capacitive and Inductive parasitic

UNIT IV CHIP PACKAGES **(9 Hrs)**

IC Assembly – Purpose, Requirements, Technologies, Wire bonding, Tape Automated Bonding, Flip Chip, Wafer Level Packaging, reliability, wafer level burn – in and test. Single chip packaging: functions, types, materials processes, properties, characteristics, trends. Multi chip packaging: types, design, comparison, trends. System – in – package (SIP); Passives: discrete, integrated, and embedded

UNIT V TESTING **(9 Hrs)**

Testing Reliability, Basic concepts, Environmental interactions. Thermal mismatch and fatigue – failures -thermo mechanically induced -electrically induced – chemically induced. Electrical Testing: System level electrical testing, Interconnection tests, Active Circuit Testing, Design for Testability

Text Books

1. Tummala, Rao R., Fundamentals of Microsystems Packaging, McGraw Hill, 2001
2. R.G. Kaduskar and V.B.Baru, Electronic Product design, Wiley India, 2011
3. Tummala, Rao R, Microelectronics packaging handbook, McGraw Hill, 2008.

Reference Books

1. Blackwell (Ed), "The electronic packaging handbook", CRC Press, 2000.
2. R.S.Khandpur, "Printed Circuit Board", Tata McGraw Hill, 2005
3. R. K. Ulrich, "Recent literature in Electronic Packaging", 2005
4. Michael L. Bushnell and Vishwani D. Agrawal, "Essentials of Electronic Testing for Digital, Memory and Mixed signal VLSI Circuits", Kluwer Academic Publishers.2000.
5. M. Abramovici, M. A. Breuer, and A.D. Friedman, "Digital System Testing and Testable Design", Computer Science Press,

Web References

1. <http://www.logopeople.in/blog/awesome-packaging-design-of-electronic-products-for-inspiration/>
2. <https://www.pinterest.com/PackagingTPI/electronic-packaging/>
3. <https://www.einfochips.com/blog/semiconductor-and-electronic-design-networks-and-profiles-to-follow-in-2018/>
4. https://en.wikipedia.org/wiki/Electronic_packaging
5. <https://nptel.ac.in/courses/108/108/108108031/>

COs /POs/PSOs Mapping

COs	Program Outcomes (POs)												Program Specific Outcomes(PSOs)		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	1	1	-	-	-	-	-	-	1	-	-			
2	3	1	1	-	-	-	-	-	-	1	-	-			
3	3	1	1	-	-	-	-	-	-	1	-	-			
4	3	1	1	-	-	-	-	-	-	1	-	-			
5	3	1	1	-	-	-	-	-	-	1	-	-			

Correlation Level: 1-Low, 2-Medium, 3- High

	PLATFORM TECHNOLOGY	L	T	P	C	Hrs
U20CSO603	(Common to EEE, ECE, ICE, MECH, CIVIL, CCE, BME, AI&DS)	3	0	0	3	45

Course Objectives

- To understand the fundamentals of developing modular application by using object oriented concepts.
- To utilize the C# and .NET framework to build distributed enterprise applications.
- To develop Console Application, Windows Application and Web Applications.
- To connect to multiple data sources and managing them effectively.
- To develop the Enterprise kind of applications

Course Outcomes

After completion of the course, the students will be able to

CO1-Understand the concept of .NET Framework.(K2)

CO2 -Develop, implement and creating Applications with C#.(K4)

CO3 -Evaluate various graphics and window forms.(K5)

CO4 - Integrating front end applications with Database connectivity.(K3)

CO5- Classifying various Enterprise applications into real world problems.(K3)

UNIT I INTRODUCTION TO .NET FRAMEWORK**(9 Hrs)**

.NET Framework - Common language Runtime (CLR) – Common Type System (CTS) – Common language Specification (CLS) – Compilation process – Assemblies – Namespaces – Command line compiler.

UNIT II C# FUNDAMENTAL**(9 Hrs)**

C# class - object - string formatting - Types - scope - Constants - C# iteration - Control flow - Operators - Array - String - Enumerations - Structures - Custom namespaces. Programming constructs – value types and reference types – object oriented concepts – Encapsulation – Inheritance – polymorphism – Interfaces – collections – Multithreading.

UNIT III GRAPHICS AND WINDOWS FORMS**(9 Hrs)**

Tool box controls – Container control – Menu – Tool bar – Tool tip Controls during design time – Run time – Graphics programming GDI+.

UNIT IV DATABASE PROGRAMMING**(9 Hrs)**

Data Access with ADO.NET – Architecture – Data reader – Data Adapter – Command – Connection – Data set – Data binding – Data Grid Control – XML based Data sets.

UNIT V J2EE**(9 Hrs)**

Enterprise Edition Overview – Multi-Tier Architecture – Best Practices – Comparison between J2EE and .NET.

Text Books

1. David Chappell, "Understanding .NET – A Tutorial and Analysis", Addison Wesley, 2002.
2. Herbert Schildt, "C# 3.0 The Complete Reference", McGraw-Hill Professional, Third Edition, 2008.
3. Keogh, "J2EE The Complete Reference", Tata McGraw-Hill, 2008.

Reference Books

1. Andrew Troelsen, "Pro C# 5.0 and the .NET 4.5 Framework", Sixth edition, A Press, 2012.
2. Joh Skeet, "C# in depth, Manning publications", Third Edition, 2014.